

Review on **MATHEMATICS AND BASIC ENGINEERING**



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Bayaw, remind yourself that



TOPICS TO BE REVIEWED

- ☐ Algebra and Advanced Math
- ☐ Trigonometry
- ☐ Geometry
- ☐ Differential Calculus
- ☐ Integral Calculus
- ☐ Differential Equation
- ☐ Basic Engineering Principles and Applications

TRY TO SOLVE THIS

What three consecutive even numbers have their sum equal to 528?

Find three consecutive odd numbers whose sum is 975.

TRY TO SOLVE THIS

What three consecutive even numbers have their sum equal to 528?

174 176 178

Find three consecutive odd numbers whose sum is 975.

323 325 327

TRY TO SOLVE THIS

The product of three consecutive integers is 6840. Find the third integer.

- A. 17
- B. 18
- C. 19
- D. 20

TRY TO SOLVE THIS

The product of three consecutive integers is 6840. Find the third integer.

- A. 17
- B. 18
- C. 19
- D. 20

$$(n - 2)(n - 1)(n) = 6840$$

ALGEBRA AND ADVANCED MATH

✓ **Binomial Theorem**

- Progression
- Work Problems
- Age Problems
- Mixture Problems
- Permutation
- Combination
- Probability

BINOMIAL THEOREM

Expansion of $(a + b)^n$

Pascal's Triangle

It is used to determine coefficients of the terms in a binomial expression.

$(a + b)^0$							1		
$(a + b)^1$				1		1			
$(a + b)^2$			1		2		1		
$(a + b)^3$		1		3		3	1		
$(a + b)^4$		1		4		6	4	1	
$(a + b)^5$	1		5		10		10	5	1

r^{th} term of $(a + b)^n$:

$$r^{th} \text{ term} = \frac{n!}{(n - r + 1)!(r - 1)!} a^{n-r+1} b^{r-1}$$

EXAMPLE:

Find the 3rd term in the expansion of $(x^2 + y)^5$.

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Using the properties of Pascal's triangle:

		1								
		1		1						
	1		2		1					
	1		3		3		1			
	1		4		6		4		1	
1		5		10		10		5		1

$$a^5b^0 + a^4b^1 + a^3b^2 + a^2b^3 + a^1b^4 + a^0b^5$$

$$a = x^2 \quad b = y$$

$$(x^2)^3y^2 = x^6y^2$$

Therefore, the 3rd term is $10x^6y^2$.

EXAMPLE:

Find the 3rd term in the expansion of $(x^2 + y)^5$.

Using the formula:

$$r^{th} \text{ term} = \frac{n!}{(n-r+1)!(r-1)!} a^{n-r+1} b^{r-1}$$

EXAMPLE:

Find the 3rd term in the expansion of $(x^2 + y)^5$.

Using the formula: $r^{th} \text{ term} = \frac{n!}{(n-r+1)!(r-1)!} a^{n-r+1} b^{r-1}$

$$r = 3$$

$$n = 5$$

$$a = x^2$$

$$b = y$$

$$\begin{aligned} 3^{rd} \text{ term} &= \frac{5!}{(5-3+1)!(3-1)!} (x^2)^{5-3+1} (y)^{3-1} \\ &= 10x^6y^2 \end{aligned}$$

TRY TO SOLVE THIS

Find the 4th term of $(2x + 3y)^5$.

- A. $3x^3 2y^2$
- B. $2x^2 3y^3$
- C. $10x^2 y^3$
- D. $720x^2 y^3$
- E. None of the above

TRY TO SOLVE THIS

Find the 4th term of $(2x + 3y)^5$.

A. $3x^3 2y^2$

B. $2x^2 3y^3$

C. $10x^2 y^3$

D. $720x^2 y^3$

E. None of the above

$$r^{th} \text{ term} = \frac{n!}{(n-r+1)!(r-1)!} a^{n-r+1} b^{r-1}$$

$$\begin{aligned} r &= 4 \\ a &= 2x \end{aligned}$$

$$\begin{aligned} n &= 5 \\ b &= 3y \end{aligned}$$

$$\begin{aligned} 4^{th} \text{ term} &= \frac{5!}{(5-4+1)!(4-1)!} (2x)^{5-4+1} (3y)^{4-1} \\ &= 10(4x^2)(27y^3) \\ &= \mathbf{1080x^2y^3} \end{aligned}$$

TRY TO SOLVE THIS

Find the third term of $(x+3y)^8$.

A. $252x^6y^2$

B. $168x^6y^2$

C. $504x^6y^2$

D. $336x^6y^2$

TRY TO SOLVE THIS

Find the third term of $(x+3y)^8$.

A. $252x^6y^2$

B. $168x^6y^2$

C. $504x^6y^2$

D. $336x^6y^2$

TRY TO SOLVE THIS

Find the third term of $(a-2b)^9$.

A. $144a^7b^2$

B. $-144a^7b^2$

C. $144a^2b^7$

D. $-144a^2b^7$

TRY TO SOLVE THIS

Find the third term of $(a-2b)^9$.

A. $144a^7b^2$

B. $-144a^7b^2$

C. $144a^2b^7$

D. $-144a^2b^7$

$$r^{th} \text{ term} = \frac{n!}{(n-r+1)!(r-1)!} a^{n-r+1} b^{r-1}$$

$$\begin{aligned} r &= 3 \\ a &= a \end{aligned}$$

$$\begin{aligned} n &= 9 \\ b &= -2b \end{aligned}$$

$$\begin{aligned} 3^{rd} \text{ term} &= \frac{9!}{(9-3+1)!(3-1)!} (a)^{9-3+1} (-2b)^{3-1} \\ &= (36)(a^7)(4b^2) \\ &= \mathbf{144a^7b^2} \end{aligned}$$

Find the third term of $(a-2b)^9$.

Using the properties of Pascal's triangle:

$(a + b)^0$						1							
$(a + b)^1$						1		1					
$(a + b)^2$						1		2		1			
$(a + b)^3$						1		3		3		1	
$(a + b)^4$						1		4		6		4	
$(a + b)^5$						1		5		10		10	
$(a + b)^6$						1		6		15		20	
$(a + b)^7$						1		7		21		35	
$(a + b)^8$						1		8		28		56	
$(a + b)^9$						1		9		36		84	

$$a^9b^0 + a^8b^1 + \boxed{a^7b^2} + a^6b^3$$

$$a = a \quad b = -2b$$

$$a^7b^2 = a^7(-2b)^2 = \mathbf{a^7(4b^2)}$$

$$= 36 (a^7(4b^2))$$

$$= \mathbf{144a^7b^2}$$

TRY TO SOLVE THIS

Find the 5th term of $(2a - 3b)^7$.

A. $-210 a^3 b^4$

B. $210 a^3 b^4$

C. $-22680 a^3 b^4$

D. $22680a^3b^4$

TRY TO SOLVE THIS

Find the 5th term of $(2a - 3b)^7$.

A. $-210 a^3 b^4$

B. $210 a^3 b^4$

C. $-22680 a^3 b^4$

D. $22680 a^3 b^4$

TRY TO SOLVE THIS

Find the 4th term of $\left(2x - \frac{1}{4}y\right)^{10}$.

ALGEBRA AND ADVANCED MATH

✓ **Binomial Theorem**

✓ **Progression**

- Work Problems
- Age Problems
- Mixture Problems

- Permutation
- Combination
- Probability

ARITHMETIC PROGRESSION

A sequence of numbers in which the **difference** of any two adjacent terms is constant.

Ex. 3, 8, 13, 18, ... ($d=5$)

n^{th} term of AP:

$$t_n = t_1 + (n - 1)d$$

$$t_6 = 3 + (6 - 1)5 = 28$$

Sum of the n terms:

$$S_n = \frac{n}{2} (t_1 + t_n)$$

$$S_6 = \frac{6}{2} (3 + 28) = 93$$

GEOMETRIC PROGRESSION

A sequence of numbers in which the **ratio** of any two adjacent terms is constant.

Ex. 2, 4, 8, 16, ... ($r = 2$)

n^{th} term of GP:

$$t_n = t_1 r^{n-1}$$

$$t_6 = (2)(2)^{6-1} = 64$$

Sum of the n terms:

$$S = \frac{t_1(r^n - 1)}{r - 1} \quad \text{when } r > 1$$

$$S = \frac{t_1(1 - r^n)}{1 - r} \quad \text{when } r < 1$$

HARMONIC PROGRESSION

A sequence of numbers in which **their reciprocals forms an arithmetic progression**

Ex. 6, 3, 2, ...

reciprocal: $\frac{1}{6}, \frac{1}{3}, \frac{1}{2}, \dots$

$$\left(d = \frac{1}{6}\right)$$

n^{th} term of HP:

$$t_n = t_1 + (n - 1)d$$

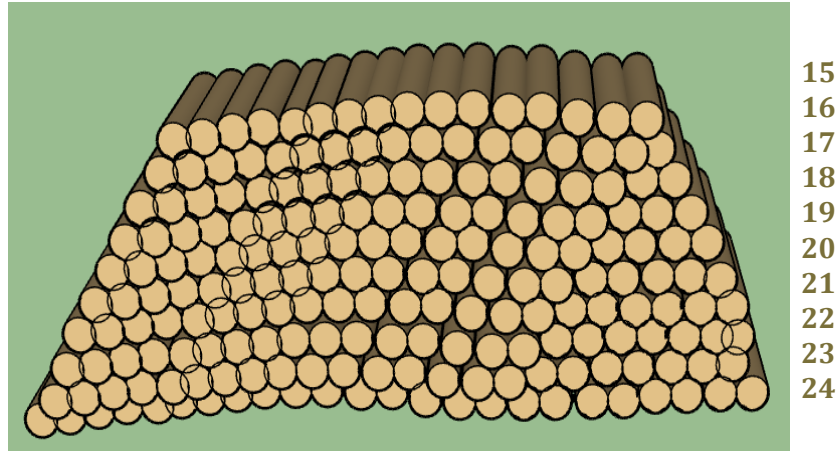
Sum of the n terms:

$$S_n = \frac{n}{2}(t_1 + t_n)$$

TRY TO SOLVE THIS

Logs are stacked in a pile with 24 logs on the bottom row and 15 on the top row. There are 10 rows in all with each row having one more log than the one above it. How many logs are in the stack?

- A. 159
- B. 160
- C. 195
- D. 199



TRY TO SOLVE THIS

Logs are stacked in a pile with 24 logs on the bottom row and 15 on the top row. There are 10 rows in all with each row having one more log than the one above it. How many logs are in the stack?

A. 159

$$15 + 16 + 17 + 18 + 19 + 20 + 21 + 22 + 23 + 24 = 195$$

B. 160

Or using the formula for the sum of n terms:

C. 195

$$S = \frac{n}{2}(t_1 + a_n) = \frac{10}{2}(15 + 24) = 195$$

D. 199

TRY TO SOLVE THIS

In a series 3,6,12,21 ... what is the 15th term?

- A. 87
- B. 168
- C. 318
- D. 411

QUADRATIC SEQUENCE

A sequence of numbers in which the **second difference** between terms is constant.

Ex. 3, 6, 12, 21, ... ($d=3$)

n^{th} term of AP:

$$t_n = an^2 + bn + c$$

TRY TO SOLVE THIS

In a series 3,6,12,21 ... what is the 15th term?

A. 87

B. 168

C. 318

D. 411

TRY TO SOLVE THIS

The population of the country increases 5% each year. Find the percentage it will increase in three years.

- A. 5%
- B. 15%
- C. 15.15%
- D. 15.76%

TRY TO SOLVE THIS

The population of the country increases 5% each year. Find the percentage it will increase in three years.

- A. 5%
- B. 15%
- C. 15.15%
- D. 15.76%**

Let 100 be the present population of the country.

Population on the first year = $100 + 0.05(100) = 100(1.05) = 105$

Population on the second year = $105(1.05) = 110.25$

Population on the third year = $110.25(1.05) = 115.7625$

Percentage increase = $115.7625 - 100 = 15.7625\%$

Another solution (Geometric Progression):

Let x be the present population of the country.

$$t_n = t_1 r^{n-1} \quad (r = 1.05)$$

$$P_3 = x(1.05)^3 = 1.157625x$$

ALGEBRA AND ADVANCED MATH

✓ **Binomial Theorem**

- Permutation

✓ **Progression**

- Combination

- Probability

✓ **Work Problems**

- Age Problems

- Mixture Problems

WORK PROBLEM

$$\text{Rate} = \frac{1}{\text{time to finish the work}}$$

$$\text{Work done} = \text{Rate} \times \text{time}$$

Time to finish the job when working together:

$$t_{\text{total}} = \frac{1}{\frac{1}{t_1} + \frac{1}{t_2} + \dots + \frac{1}{t_n}}$$

EXAMPLE:

If Marites can do a job in 4 hours and Marisol can do the same job in 8 hours, working together from the start,

(a) what part of the job have they done in 2 hours, and

(b) how many hours can they finish the job?

SOLUTION:

Work done in 2 hours:

$$\frac{1}{4}(2) + \frac{1}{8}(2) = \frac{3}{4}$$

Time to finish the job:

$$\begin{aligned} & \frac{1}{\frac{1}{4} + \frac{1}{8}} \\ &= 2.\overline{666} \text{ hours} \\ &= \mathbf{2 \text{ hours and } 40 \text{ minutes}} \end{aligned}$$

TRY TO SOLVE THIS

A pipe can fill a swimming pool in 10 hours. If a second pipe is open, the two pipes together can fill the pool in 4 hours. How long will it take the second pipe alone to fill the pool?

- A. Exactly 5 hours
- B. 5 hours and 30 minutes
- C. 6 hours and 7 minutes
- D. 6 hours and 40 minutes



TRY TO SOLVE THIS

A pipe can fill a swimming pool in 10 hours. If a second pipe is open, the two pipes together can fill the pool in 4 hours. How long will it take the second pipe alone to fill the pool?

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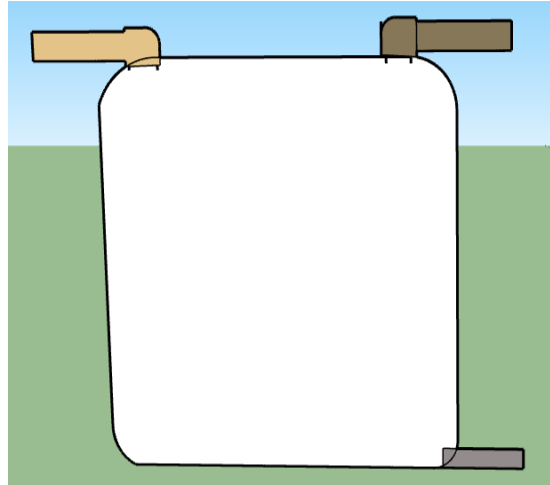
$$\frac{1}{\frac{1}{10} + \frac{1}{x}} = 4$$

$$x = 6.6667 \text{ hours}$$

TRY TO SOLVE THIS

One pipe can fill a tank in 5 hours and another pipe can fill the same tank in 6 hours. A drainpipe can empty the full content of the tank in 15 hours. With all the three pipes open, how long will it take to fill the tank?

- A. 2 hours and 20 minutes
- B. 2 hours and 30 minutes
- C. 3 hours and 30 minutes
- D. 3 hours and 20 minutes



TRY TO SOLVE THIS

One pipe can fill a tank in 5 hours and another pipe can fill the same tank in 6 hours. A drainpipe can empty the full content of the tank in 15 hours. With all the three pipes open, how long will it take to fill the tank?

- A. 2 hours and 20 minutes
- B. 2 hours and 30 minutes
- C. 3 hours and 30 minutes
- D. 3 hours and 20 minutes**

$$\frac{1}{\frac{1}{5} + \frac{1}{6} - \frac{1}{15}} = 3.3333 \text{ hours}$$

ALGEBRA AND ADVANCED MATH

- ✓ **Binomial Theorem**
 - Mixture Problems
- ✓ **Progression**
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- ✓ **Work Problems**
 - Probability
- ✓ **Age Problems**

PRINCIPLES OF AGE PROBLEMS

The difference of the ages of two persons is constant.

If x is the age of Bongbong now:

His age 5 years ago is $x - 5$.

His age in 7 years is $x + 7$.

EXAMPLE:

Two times the father's age is 8 more than six times his sons age. Ten years ago, the sum of their ages was 44. The age of the son is _____.

Let x = father's age; y = son's age

Two times the father's age is 8 more than six times his sons age.

$$2x = 8 + 6y$$

$$x = \frac{8 + 6y}{2}$$

Ten years ago, the sum of their ages was 44.

$$(x - 10) + (y - 10) = 44$$

$$\left[\left(\frac{8 + 6y}{2} \right) - 10 \right] + (y - 10) = 44$$

$$y = 15$$

*The age of the son is **15**.*

TRY TO SOLVE THIS

A father is three times as old as his son. Four years ago, he was four times as old as his son was at that time. How old is his son now?

- A. 36 years
- B. 24 years
- C. 32 years
- D. 12 years

TRY TO SOLVE THIS

A father is three times as old as his son. Four years ago, he was four times as old as his son was at that time. How old is his son now?

$$x = 3y$$

$$(x - 4) = 4(y - 4)$$

$$((3y) - 4) = 4(y - 4)$$

$$y = 12$$

- A. 36 years
- B. 24 years
- C. 32 years
- D. 12 years**

TRY TO SOLVE THIS

Ferdinand is 11 years old today. If her favorite aunt is 31 years old, how many years from now will the aunt be twice as old as he?

- A. 7
- B. 8
- C. 9
- D. 10

TRY TO SOLVE THIS

Ferdinand is 11 years old today. If her favorite aunt is 31 years old, how many years from now will the aunt be twice as old as he?

A. 7

B. 8

C. 9

D. 10

$$(31 + x) = 2(11 + x)$$

$$x = 9$$

TRY TO SOLVE THIS

The ages of the mother and her daughter are 45 and 5 years, respectively. How many years will the mother be three times as old as her daughter?

- A. 5
- B. 10
- C. 15
- D. 20

TRY TO SOLVE THIS

The ages of the mother and her daughter are 45 and 5 years, respectively. How many years will the mother be three times as old as her daughter?

- A. 5
- B. 10
- C. 15**
- D. 20

ALGEBRA AND ADVANCED MATH

✓ **Binomial Theorem**

✓ **Progression**

✓ **Work Problems**

✓ **Age Problems**

✓ **Mixture Problems**

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MIXTURE PROBLEM

EXAMPLE:

How many grams of gold must be added with 500 grams of an alloy containing 30% gold and 70% silver in order to produce another alloy containing 40% gold and 60% silver?

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500		x		500 + x
30% gold 70% silver	+	100% gold	=	40% gold 60% silver
500(0.30)	+	x(1)	=	(500 + x)(0.40)
<i>x = 83.33 grams</i>				

MIXTURE PROBLEM

EXAMPLE:

How many liters of a 15% solution of alcohol should be added to 3 liters of the 30% solution to get a 20 percent solution?

MIXTURE PROBLEM

EXAMPLE:

How many liters of a 15% solution of alcohol should be added to 3 liters of the 30% solution to get a 20 percent solution?

3		x		3 + x
30% solution	+	15% solution	=	20% solution
$3(0.30)$	+	$x(0.15)$	=	$(3 + x)(0.20)$
$x = 6 \text{ liters}$				

TRY TO SOLVE THIS

How many kilograms of pure water is to be added to 100 kilograms of a 30% saline solution to make it a 10% saline solution?

- A. 200 kilograms
- B. 180 kilograms
- C. 150 kilograms
- D. 100 kilograms

TRY TO SOLVE THIS

How many kilograms of pure water is to be added to 100 kilograms of a 30% saline solution to make it a 10% saline solution?

- A. 200 kilograms**
- B. 180 kilograms
- C. 150 kilograms
- D. 100 kilograms

TRY TO SOLVE THIS

Determine how many liters of an 8% acid solution and how many liters of a 10% acid solution should be mixed to obtain 10 liters of 9% acid solution.

- A. 5 liters of each solution
- B. 6 liters of the 8%; 4 liters of the 10%
- C. 4 liters of the 8%; 6 liters of the 10%
- D. None of the above

TRY TO SOLVE THIS

Determine how many liters of an 8% acid solution and how many liters of a 10% acid solution should be mixed to obtain 10 liters of 9% acid solution.

- A. 5 liters of each solution**
- B. 6 liters of the 8%; 4 liters of the 10%
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- D. None of the above

ALGEBRA AND ADVANCED MATH

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✓ **Permutation**

- Combination
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PERMUTATION

This refers to an arrangement of objects **in definite order**.

Permutation of n **different things** taken r at a time

$$P(n, r) = \frac{n!}{(n - r)!} \quad \text{and} \quad P(n, n) = n!$$

EXAMPLE: *How many permutations can be made out of the letters in the word **ABE**?*

$$P(3,3) = 3! = \mathbf{6\ ways}$$

(ABE, AEB, BEA, BAE, EAB, EBA)

EXAMPLE: How many permutations can be made out of the letters in the word **ABE** taken 2 at a time?

$$P(n, r) = \frac{3!}{(3-2)!} = \mathbf{6 \text{ ways}}$$

(AB, AE, BA, BE, EA, EB)

Permutation of n objects with **some objects identical**

$$P = \frac{n!}{q! r! \dots}$$

EXAMPLE: How many permutations can be made out of the letters in the word **ENGINEER**?

$$n=8; (3-E, 2-N)$$

$$P = \frac{8!}{3! 2!} = \mathbf{3,360 \text{ ways}}$$

Permutation of n Things in a **circle** is: $P = (n - 1)!$

EXAMPLE: How many ways can 7 people be seated at a round table?

$$P = (7 - 1)! = \mathbf{720 \text{ ways}}$$

TRY TO SOLVE THIS

How many permutations can be made out of the letters in the word **BAYAW**?

60 ways

ALGEBRA AND ADVANCED MATH

- ✓ **Binomial Theorem**
- ✓ **Mixture Problems**
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COMBINATION

This refers to a collection of objects **without regard to sequence or order** in which they were chosen.

$$C(n, r) = \frac{n!}{(n - r)! r!}$$

EXAMPLE: How many ways can you choose 2 letters from the word **ABE**?

$$C(3, 2) = \frac{3!}{(3-2)!2!} = \mathbf{3 \text{ ways}}$$

$$(AB=BA, AE=EA, BE=EB)$$

EXAMPLE: How many ways can you draw 3 QUEENS and 2 KINGS from a deck of 52 cards?

$$C(4,3) = \frac{4!}{(4-3)! 3!} = 4 \text{ ways}$$

$$C(4,2) = \frac{4!}{(4-2)! 2!} = 6 \text{ ways}$$

$$\mathbf{C = 4 \times 6 = 24 \text{ ways}}$$

TRY TO SOLVE THIS

How many ways can a committee of five may be selected from an organization with 35 members?

324632 ways

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PROBABILITY

It is the likelihood that an event will occur expressed as the ratio of the number of favorable outcomes in the set of outcomes divided by the total number of possible outcomes.

$$\text{Probability} = \frac{\text{Number of Favorable Ways}}{\text{Total Number of ways}}$$

EXAMPLE: If a card is drawn out of an ordinary deck of cards, find the probability that it is the heart?



$$\text{Probability} = \frac{13}{52} = 0.25 \text{ or } 25\%$$

TRY TO SOLVE THIS

What is the probability of drawing queen of hearts on a deck of cards?

A. $\frac{4}{52}$

B. $\frac{1}{13}$

C. $\frac{1}{52}$

D. $\frac{4}{13}$

TRY TO SOLVE THIS

What is the probability of drawing queen of hearts on a deck of cards?

A. $4/52$

B. $1/13$

C. $1/52$

D. $4/13$



TRIGONOMETRY

- Functions of Right Triangle
- Oblique Triangles

FUNCTIONS OF RIGHT TRIANGLE

$$\sin\theta = \frac{\text{opposite side}}{\text{hypotenuse}} \quad (\text{soh})$$

$$\csc\theta = \frac{\text{hypotenuse}}{\text{opposite side}} \quad (\text{cho})$$

$$\cos\theta = \frac{\text{adjacent side}}{\text{hypotenuse}} \quad (\text{cah})$$

$$\sec\theta = \frac{\text{hypotenuse}}{\text{adjacent side}} \quad (\text{sha})$$

$$\tan\theta = \frac{\text{opposite side}}{\text{adjacent side}} \quad (\text{toa})$$

$$\cot\theta = \frac{\text{adjacent side}}{\text{opposite side}} \quad (\text{cao})$$

PYTHAGOREAN THEOREM

$$c^2 = a^2 + b^2$$

OBLIQUE TRIANGLES

SINE LAW

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

COSINE LAW

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

TRY TO SOLVE THIS

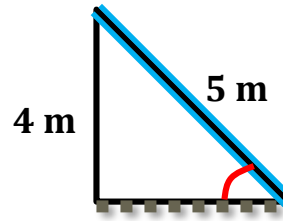
A five-meter-long ladder leans against a wall, with the top of the ladder being four meters above the ground. What is the approximate angle that the ladder makes with the ground?

- A. 35 degrees
- B. 43 degrees
- C. 53 degrees
- D. 55 degree

TRY TO SOLVE THIS

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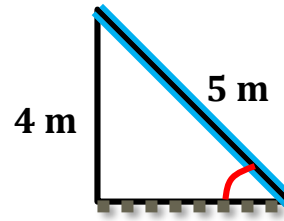
- A. 35 degrees
- B. 43 degrees
- C. 53 degrees
- D. 55 degree



TRY TO SOLVE THIS

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- A. 35 degrees
- B. 43 degrees
- C. 53 degrees**
- D. 55 degree



$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\theta = \sin^{-1} \left(\frac{4}{5} \right)$$

$$\theta = 53.13^\circ$$

TRY TO SOLVE THIS

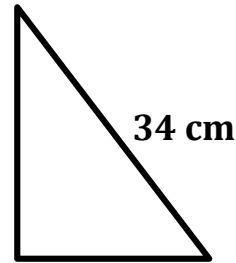
The hypotenuse of a right triangle is 34 cm. Find the length of the shortest leg if it is 14 cm shorter than the other leg.

- A. 15 cm
- B. 16 cm
- C. 17 cm
- D. 18 cm

TRY TO SOLVE THIS

The hypotenuse of a right triangle is 34 cm. Find the length of the shortest leg if it is 14 cm shorter than the other leg.

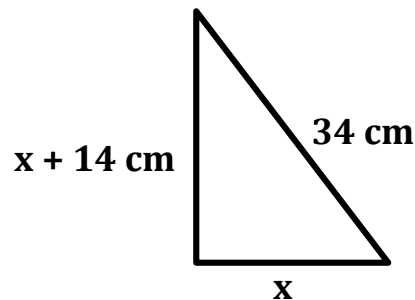
- A. 15 cm
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TRY TO SOLVE THIS

The hypotenuse of a right triangle is 34 cm. Find the length of the shortest leg if it is 14 cm shorter than the other leg.

- A. 15 cm
- B. 16 cm**
- C. 17 cm
- D. 18 cm



$$c^2 = a^2 + b^2$$

$$34^2 = x^2 + (x + 14)^2$$

$$x = 16$$

TRY TO SOLVE THIS

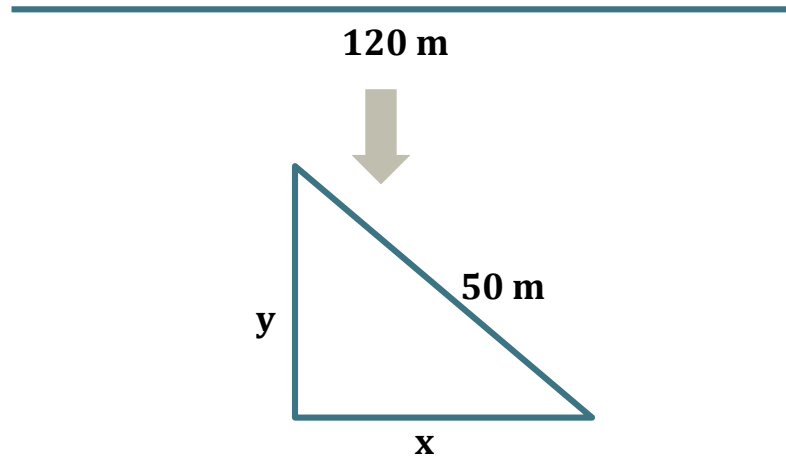
A piece of wire 120 meters long is bent to form a right triangle with a hypotenuse 50 meters long. Find the length of the other two sides of the triangle.

- A. 50 m and 20 m
- B. 45m and 25m
- C. 35 meters
- D. 30 m and 40 m

TRY TO SOLVE THIS

A piece of wire 120 meters long is bent to form a right triangle with a hypotenuse 50 meters long. Find the length of the other two sides of the triangle.

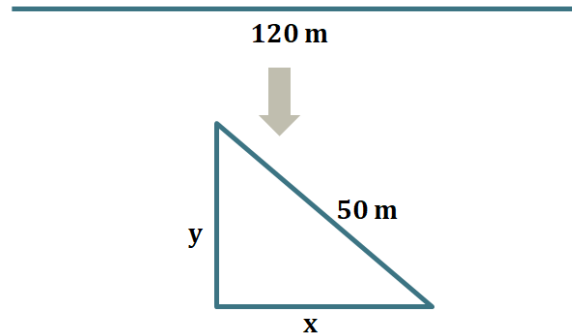
- A. 50 m and 20 m
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- D. 30 m and 40 m



TRY TO SOLVE THIS

A piece of wire 120 meters long is bent to form a right triangle with a hypotenuse 50 meters long. Find the length of the other two sides of the triangle.

- A. 50 m and 20 m
- B. 45m and 25m
- C. 35 meters
- D. 30 m and 40 m**



$$x + y + 50 = 120$$

$$x = 70 - y$$

$$x^2 + y^2 = 50^2$$

$$(70 - y)^2 + y^2 = 50^2$$

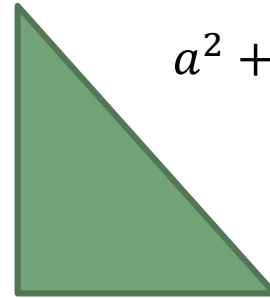
$$y = 30$$

$$x = 70 - 30 = 40$$

TRY TO SOLVE THIS

The sides of a right triangle form an arithmetic progression with a common difference of 6. Find the sides of the triangle .

- A. 6, 12, 18
- B. 9, 15, 21
- C. 10, 16, 22
- D. 18, 24, 30

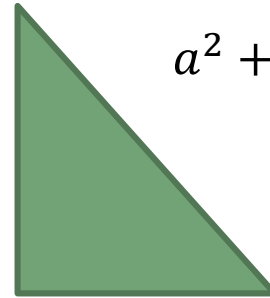


$$a^2 + b^2 = c^2$$

TRY TO SOLVE THIS

The sides of a right triangle form an arithmetic progression with a common difference of 6. Find the sides of the triangle .

- A. 6, 12, 18
- B. 9, 15, 21
- C. 10, 16, 22
- D. 18, 24, 30**



$$a^2 + b^2 = c^2$$

TRY TO SOLVE THIS

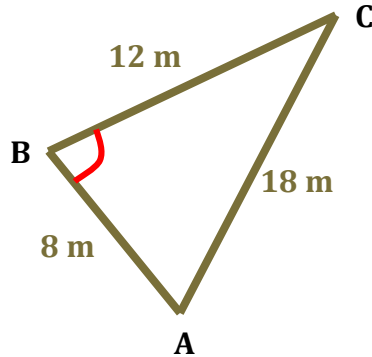
A triangular plot of land is to be used as a vegetable garden. The measures of the plot are $AB = 8$ meters, $BC = 12$ meters and $AC = 18$ meters. Find the measure of angle B.

- A. 127.2°
- B. 20.7°
- C. 90°
- D. 70.2°

TRY TO SOLVE THIS

A triangular plot of land is to be used as a vegetable garden. The measures of the plot are $AB = 8$ meters, $BC = 12$ meters and $AC = 18$ meters. Find the measure of angle B.

- A. 127.2°
- B. 20.7°
- C. 90°
- D. 70.2°



TRY TO SOLVE THIS

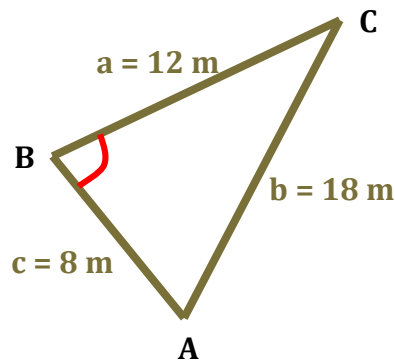
A triangular plot of land is to be used as a vegetable garden. The measures of the plot are AB = 8 meters, BC = 12 meters and AC = 18 meters. Find the measure of angle B.

A. **127.2°**

B. 20.7°

C. 90°

D. 70.2°



$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$18^2 = 12^2 + 8^2 - 2(12)(8) \cos B$$

$$324 = 144 + 64 - 192 \cos B$$

$$192 \cos B = -116$$

$$B = \cos^{-1} \left(-\frac{116}{192} \right)$$

$$\mathbf{B = 127.2^\circ}$$

TRY TO SOLVE THIS

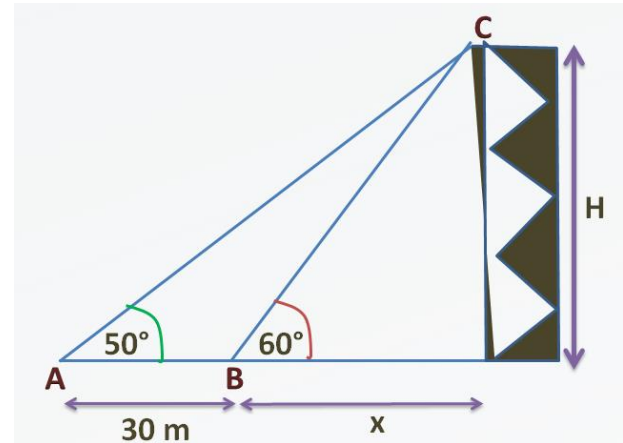
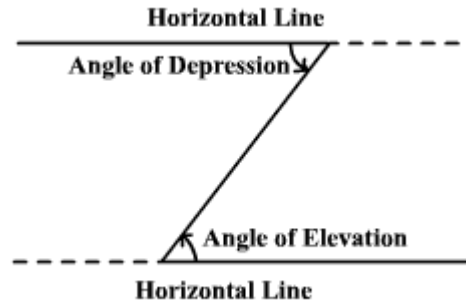
The **angle of elevation** to the top C of a building from two points A and B on level ground are 50 degrees and 60 degrees respectively. The distance between points A and B is 30 meters. Find the approximate height H of the building.

- A. 95 meters
- B. 100 meters
- C. 110 meters
- D. 115 meters

TRY TO SOLVE THIS

The **angle of elevation** to the top C of a building from two points A and B on level ground are 50 degrees and 60 degrees respectively. The distance between points A and B is 30 meters. Find the approximate height H of the building.

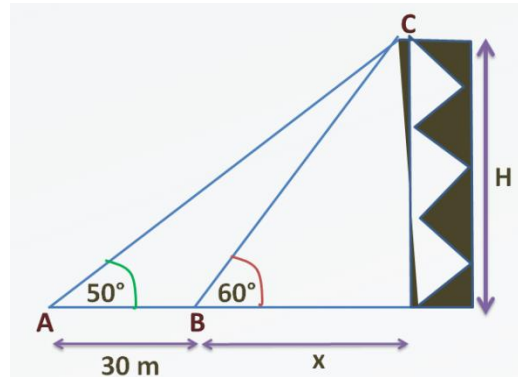
- A. 95 meters
- B. 100 meters
- C. 110 meters
- D. 115 meters



TRY TO SOLVE THIS

The angle of elevation to the top C of a building from two points A and B on level ground are 50 degrees and 60 degrees respectively. The distance between points A and B is 30 meters. Find the approximate height H of the building.

- A. 95 meters
- B. 100 meters
- C. 110 meters
- D. 115 meters**



$$\tan 60 = \frac{H}{x}$$

$$x = \frac{H}{\tan 60}$$

$$\tan 50 = \frac{H}{30 + x}$$

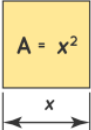
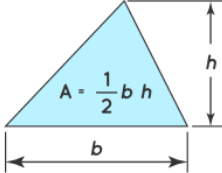
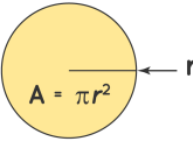
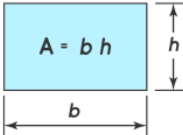
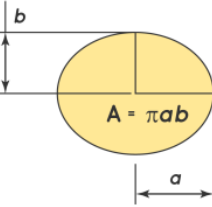
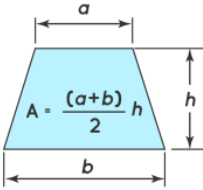
$$\tan 50 = \frac{H}{30 + \left(\frac{H}{\tan 60}\right)}$$

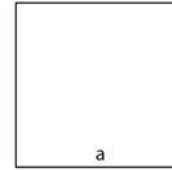
$$H \approx 115 \text{ meters}$$

GEOMETRY

- Plane Geometry
- Solid Geometry
- Analytic Geometry

AREA AND PERIMETER FORMULAS

Square  $A = x^2$	Triangle  $A = \frac{1}{2} b h$	Circle  $A = \pi r^2$
Rectangle  $A = b h$	Eclipse  $A = \pi a b$	Trapezoid  $A = \frac{(a+b)}{2} h$



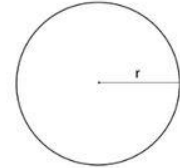
Square

$P = 4a$



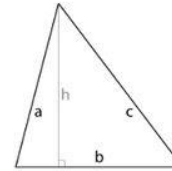
Rectangle

$P = 2l + 2w$



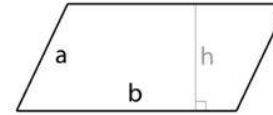
Circle

$P = 2\pi r$



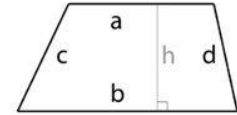
Triangle

$P = a + b + c$



Parallelogram

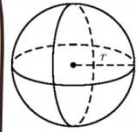
$P = 2a + 2b$



Trapezoid

$P = a + b + c + d$

SURFACE AREA AND VOLUME FORMULAS



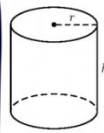
Volume

$$V = \frac{4}{3} \pi r^3$$

Surface Area

$$S = 4\pi r^2$$

Sphere



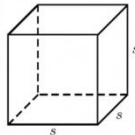
Volume

$$V = \pi r^2 h$$

Surface Area

$$S = 2\pi r h + 2\pi r^2$$

Circular Cylinder



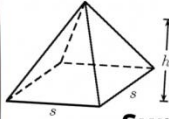
Volume

$$V = s^3$$

Surface Area

$$S = 6s^2$$

Cube



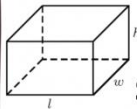
Volume

$$V = \frac{1}{3} s^2 h$$

Surface Area

$$S = s(s + \sqrt{s^2 + 4h^2})$$

Square Pyramid



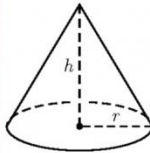
Volume

$$V = lwh$$

Surface Area

$$S = 2lw + 2lh + 2wh$$

Rectangular Solid



Volume

$$V = \frac{1}{3} \pi r^2 h$$

Surface Area

$$S = \pi r \sqrt{r^2 + h^2} + \pi r^2$$

Circular Cone

TRY TO SOLVE THIS

The length of the side of a square is increased by 100%. Its perimeter is increased by:

- A. 25%
- B. 100%
- C. 200%
- D. 300%

TRY TO SOLVE THIS

The length of the side of a square is increased by 100%. Its perimeter is increased by:

- A. 25%
- B. 100%**
- C. 200%
- D. 300%

Let 1m be the length of the side of the square.

$$\text{Perimeter} = 4(1) = 4\text{m}$$

If increased to 2m,

$$\text{Perimeter} = 4(2) = 8\text{m}$$

$$\text{Perimeter increase} = 8 - 4 = 4\text{m or}$$

$$\frac{4}{4} = 100\% \text{ from the original perimeter}$$

TRY TO SOLVE THIS

Each side of a cube is increased by 10%. By what percent is the volume of the cube increased?

- A. 23.4%
- B. 33.1%
- C. 3%
- D. 34.56%

TRY TO SOLVE THIS

Each side of a cube is increased by 10%. By what percent is the volume of the cube increased?

- A. 23.4%
- B. 33.1%**
- C. 3%
- D. 34.56%

Let 1m be the measurement of the side of the cube.

$$\text{Volume} = s^3 = 1^3 = 1m^3$$

If increased to 10% or 1.1 m,

$$\text{Volume} = s^3 = 1.1^3 = 1.331m^3$$

$$\text{Volume increase} = 1.331 - 1 = 0.331 m^3 \text{ or}$$

$$\frac{0.331}{1} = 33.1\% \text{ from the original volume}$$

TRY TO SOLVE THIS

If the edge of the cube is increased by 30%, by how much is the surface area increased?

- A. 63%
- B. 65%
- C. 67%
- D. 69%

TRY TO SOLVE THIS

If the edge of the cube is increased by 30%, by how much is the surface area increased?

- A. 63%
- B. 65%
- C. 67%
- D. 69%**

Let 1m be the measurement of the side of the cube.

$$SA = 6s^2 = 6(1)^2 = 6m^2$$

If increased to 30% or 1.3 m,

$$SA = 6s^2 = 6(1.3)^2 = 10.14m^2$$

$$SA \text{ increase} = 10.14 - 6 = 4.14m^2$$

$$\frac{4.14}{6} = 69\% \text{ from the original SA}$$

DISTANCE BETWEEN TWO POINTS

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

GENERAL EQUATION OF A LINE

$$Ax + By + C = 0$$

SLOPE OF THE LINE

$$\text{Slope, } m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

STANDARD EQUATIONS OF LINES

Point-slope form:

$$y - y_1 = m(x - x_1)$$

Slope-intercept form:

$$y = mx + b$$

Intercept form:

$$\frac{x}{a} = \frac{y}{b} = 1$$

Two-point form:

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1} = m$$

Note: All of these four standard equations can be reduced into the point-slope form.

TRY TO SOLVE THIS

The midpoint of $(-2,9)$ and a point S is given to be $(2,1)$. Which ordered pair corresponds to S ?

- A. $(2,11)$
- B. $(6,11)$
- C. $(2,-7)$
- D. $(6,-7)$

TRY TO SOLVE THIS

The midpoint of $(-2,9)$ and a point S is given to be $(2,1)$. Which ordered pair corresponds to S ?

- A. $(2,11)$
- B. $(6,11)$
- C. $(2,-7)$
- D. $(6,-7)$**

TRY TO SOLVE THIS

A circle with center at $(1,1)$ contains the point $(-3,4)$. What is the length of the diameter?

- A. 5
- B. 7
- C. 8
- D. 10

TRY TO SOLVE THIS

A circle with center at $(1,1)$ contains the point $(-3,4)$. What is the length of the diameter?

- A. 5
- B. 7
- C. 8
- D. 10**

TRY TO SOLVE THIS

Find the equation of the line passing through $(4, 0)$ and is parallel to the line $x - y - 2 = 0$.

A. $y + x + 4 = 0$

B. $y - x + 4 = 0$

C. $y - x - 4 = 0$

D. $y + x - 4 = 0$

TRY TO SOLVE THIS

Find the equation of the line passing through $(4, 0)$ and is parallel to the line $x - y - 2 = 0$.

A. $y + x + 4 = 0$

B. $y - x + 4 = 0$

C. $y - x - 4 = 0$

D. $y + x - 4 = 0$

$$x - y - 2 = 0$$

$$y = x - 2$$

$$m=1$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 1(x - 4)$$

$$y = x - 4$$

$$\mathbf{y - x + 4 = 0}$$

TRY TO SOLVE THIS

Find the equation of the line passing through $(4, 0)$ and is perpendicular to the line $x - y - 2 = 0$.

A. $y + x + 4 = 0$

B. $y - x + 4 = 0$

C. $y - x - 4 = 0$

D. $y + x - 4 = 0$

TRY TO SOLVE THIS

Find the equation of the line passing through $(4, 0)$ and is perpendicular to the line $x - y - 2 = 0$.

A. $y + x + 4 = 0$

B. $y - x + 4 = 0$

C. $y - x - 4 = 0$

D. $y + x - 4 = 0$

$$x - y - 2 = 0$$

$$y = x - 2$$

$$m=1$$

Use the negative reciprocal

$$m = -1$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = -1(x - 4)$$

$$y = -x + 4$$

$$\mathbf{y + x - 4 = 0}$$

TRY TO SOLVE THIS

Find the equation of the line passing through $(6, -3)$ and $(2, 0)$.

- A. $4x - 3y = 6$
- B. $4x + 3y = 6$
- C. $3x - 4y = 6$
- D. $3x + 4y = 6$

TRY TO SOLVE THIS

Find the equation of the line passing through $(6, -3)$ and $(2, 0)$.

A. $4x - 3y = 6$

B. $4x + 3y = 6$

C. $3x - 4y = 6$

D. $3x + 4y = 6$

$$(y - y_1) = m(x - x_1)$$

$$m = \frac{(y_2 - y_1)}{(x_2 - x_1)} = \frac{0 - (-3)}{2 - 6} = \frac{3}{-4}$$

$$y - (-3) = \frac{3}{-4}(x - 6)$$

$$-4 \left[y + 3 = \frac{3x - 18}{-4} \right]$$

$$-4y - 12 = 3x - 18$$

$$\mathbf{3x + 4y = 6}$$

CONIC SECTIONS

These sections are formed if a **plane is made to pass through a cone**.



parabola



circle



ellipse



hyperbola

STANDARD EQUATIONS

Center at (h, k)

CIRCLE

$$(x - h)^2 + (y - k)^2 = r^2$$

STANDARD EQUATIONS

Center at (h, k)

PARABOLA

$$(y - k)^2 = 4a(x - h)^2$$

$$(y - k)^2 = -4a(x - h)^2$$

$$(x - h)^2 = 4a(y - k)^2$$

$$(x - h)^2 = -4a(y - k)^2$$

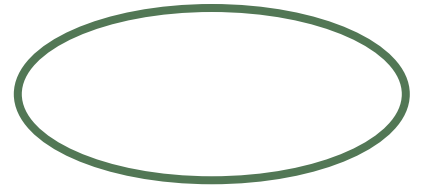


STANDARD EQUATIONS

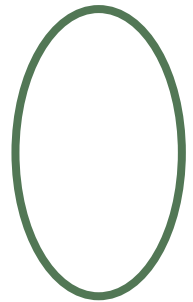
Center at (h, k)

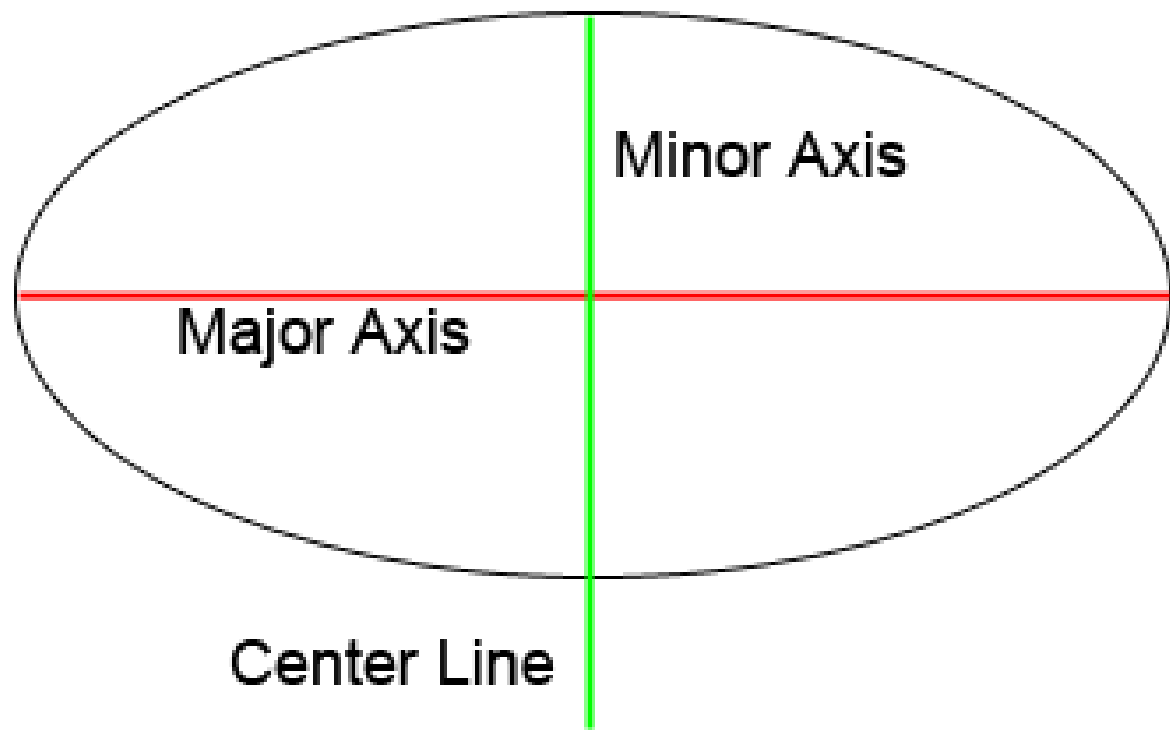
ELLIPSE

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$$



$$\frac{(y - k)^2}{a^2} + \frac{(x - h)^2}{b^2} = 1$$



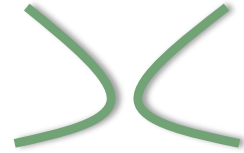


STANDARD EQUATIONS

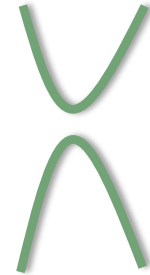
Center at (h, k)

HYPERBOLA

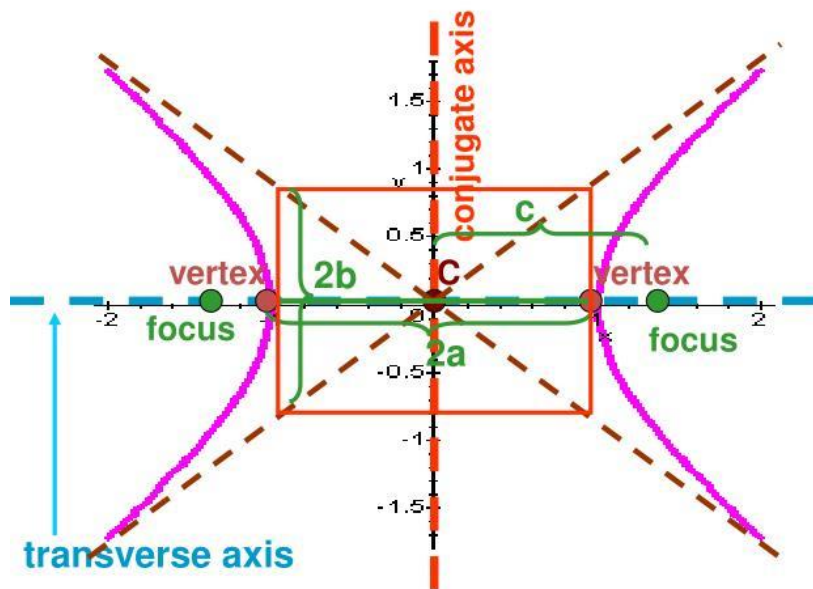
$$\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1$$



$$\frac{(y - k)^2}{a^2} - \frac{(x - h)^2}{b^2} = 1$$



PARTS OF A HYPERBOLA



c is the distance from the center to the vertex.

For a hyperbola: $c^2 = a^2 + b^2$

The distance between the vertices is **2a**

Midpoint of the line joining the vertices is the center (**C**) of the hyperbola

Construct a rectangle with its center at **C** and length **2a**

The width of this rectangle is **2b** the conjugate axis length.

The diagonals of this rectangle are asymptotes of the segments of the hyperbola.

STANDARD EQUATIONS

Center at (h, k)

CIRCLE

$$(x - h)^2 + (y - k)^2 = r^2$$

EXAMPLE:

Find the center and radius of the circle

$$x^2 + y^2 - 6x - 8y + 9 = 0.$$

Standard equation of a circle: $(x - h)^2 + (y - k)^2 = r^2$

EXAMPLE:

Find the center and radius of the circle $x^2 + y^2 - 6x - 8y + 9 = 0$.

Standard equation of a circle: $(x - h)^2 + (y - k)^2 = r^2$

$$x^2 + y^2 - 6x - 8y + 9 = 0$$

$$x^2 - 6x + y^2 - 8y = -9$$

$$(x^2 - 6x + \underline{\quad}) + (y^2 - 8y + \underline{\quad}) = -9 + \underline{\quad} + \underline{\quad}$$

$$(x^2 - 6x + 9) + (y^2 - 8y + 16) = -9 + 9 + 16$$

$$(x - 3)^2 + (y - 4)^2 = 4^2$$

$$h = 3, k = 4, r = 4$$

Therefore, the center is at (3, 4) and the radius is 4.

TRY TO SOLVE THIS

Determine the radius of the circle whose equation

$$x^2 + y^2 - 12x + 20y + 15 = 0.$$

- A. 121
- B. 100
- C. 10
- D. 11

TRY TO SOLVE THIS

Determine the radius of the circle whose equation

$$x^2 + y^2 - 12x + 20y + 15 = 0.$$

A. 121

B. 100

C. 10

D. 11

Standard equation of a circle:

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 - 12x + 20y + 15 = 0$$

$$x^2 - 12x + y^2 + 20y = -15$$

$$(x^2 - 12x + 36) + (y^2 + 20y + 100) = -15 + 36 + 100$$

$$(x - 6)^2 + (y + 10)^2 = 121$$

$$(x - 6)^2 + (y + 10)^2 = 11^2$$

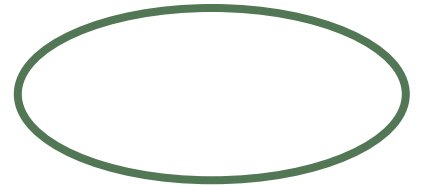
$$**r = 11**$$

STANDARD EQUATIONS

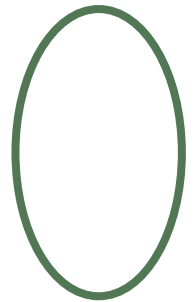
Center at (h, k)

ELLIPSE

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$$



$$\frac{(y - k)^2}{a^2} + \frac{(x - h)^2}{b^2} = 1$$



EXAMPLE:

Find the center and the length of the major axis of an ellipse with equation

$$x^2 + 6x + 4y^2 + 16y + 9 = 0$$

Standard equation of an ellipse: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

EXAMPLE:

Find the center and the length of the major axis of an ellipse with equation

$$x^2 + 6x + 4y^2 + 16y + 9 = 0$$

Standard equation of an ellipse: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

$$x^2 + 6x + 4y^2 + 16y + 9 = 0$$

$$x^2 + 6x + 4y^2 + 16y = -9$$

$$(x^2 + 6x) + 4(y^2 + 4y) = -9$$

$$(x^2 + 6x + 9) + 4(y^2 + 4y + 4)$$

$$= -9 + 9 + (4 \times 4)$$

$$(x + 3)^2 + 4(y + 2)^2 = 16$$

$$\frac{(x + 3)^2 + 4(y + 2)^2}{16} = \frac{16}{16}$$

$$\frac{(x + 3)^2}{16} + \frac{(y + 2)^2}{4} = 1$$

$$\frac{(x - (-3))^2}{4^2} + \frac{(y - (-2))^2}{2^2} = 1$$

$$h = -3, k = -2, a = 4$$

Therefore, the center is at (-3, -2), the length of the major axis is $2a = 2(4) = 8$.

TRY TO SOLVE THIS

The length of the major axis of the ellipse with equation

$$4x^2 + 9y^2 = 36 \text{ is}$$

- A. 3
- B. 2
- C. 4
- D. 6

TRY TO SOLVE THIS

The length of the major axis of the ellipse with equation

$$4x^2 + 9y^2 = 36 \text{ is}$$

A. 3

B. 2

C. 4

D. 6

Standard equation of an ellipse with center at (0, 0):

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$4x^2 + 9y^2 = 36$$

$$\frac{4x^2 + 9y^2 = 36}{36}$$

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$\frac{x^2}{3^2} + \frac{y^2}{2^2} = 1$$

$$a = 3$$

Thus, length of major axis is $2a = 2(3) = 6$.

STANDARD EQUATIONS

Center at (h, k)

PARABOLA

$$(y - k)^2 = 4a(x - h)^2$$

$$(y - k)^2 = -4a(x - h)^2$$

$$(x - h)^2 = 4a(y - k)^2$$

$$(x - h)^2 = -4a(y - k)^2$$



TRY TO SOLVE THIS

The vertex of the parabola $x^2 - 4x - 8y - 28 = 0$ is at:

- A. $(2, 4)$
- B. $(2, -4)$
- C. $(4, 2)$
- D. $(-4, 2)$

TRY TO SOLVE THIS

The vertex of the parabola $x^2 - 4x - 8y - 28 = 0$ is at:

A. $(2, 4)$

B. $(2, -4)$

C. $(4, 2)$

D. $(-4, 2)$

Standard equation of a parabola with vertex at (h, k) :

$$(x - h)^2 = 4a(y - k)$$

$$x^2 - 4x - 8y - 28 = 0$$

$$x^2 - 4x = 8y + 28$$

$$(x^2 - 4x + 4) = 8y + 28 + 4$$

$$(x - 2)^2 = 8y + 32$$

$$(x - 2)^2 = 8(y + 4)$$

$$(x - 2)^2 = 8(y - (-4))$$

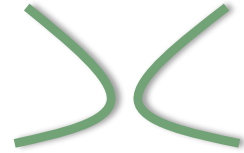
Thus, the vertex is $(2, -4)$.

STANDARD EQUATIONS

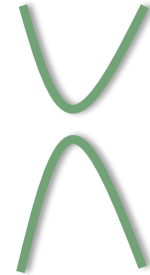
Center at (h, k)

HYPERBOLA

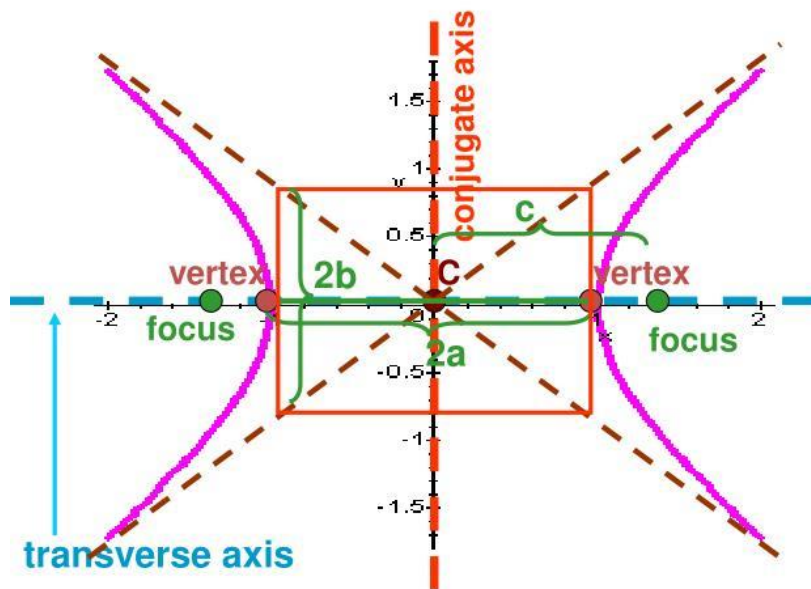
$$\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1$$



$$\frac{(y - k)^2}{a^2} - \frac{(x - h)^2}{b^2} = 1$$



PARTS OF A HYPERBOLA



c is the distance from the center to the vertex.

For a hyperbola: $c^2 = a^2 + b^2$

The distance between the vertices is **2a**

Midpoint of the line joining the vertices is the center (**C**) of the hyperbola

Construct a rectangle with its center at **C** and length **2a**

The width of this rectangle is **2b** the conjugate axis length.

The diagonals of this rectangle are asymptotes of the segments of the hyperbola.

TRY TO SOLVE THIS

Find the center of the hyperbola
 $16y^2 - 9x^2 - 64y - 80 = 0$.

- A. $(2, 0)$
- B. $(0, 2)$
- C. $(0, -2)$
- D. $(-2, 0)$

TRY TO SOLVE THIS

Find the center of the hyperbola
 $16y^2 - 9x^2 - 64y - 80 = 0$.

- A. $(2, 0)$
- B. $(0, 2)$**
- C. $(0, -2)$
- D. $(-2, 0)$

Standard equation of a hyperbola with center at (h, k) :

$$\frac{(y - k)^2}{a^2} - \frac{(x - h)^2}{b^2} = 1$$

$$16y^2 - 9x^2 - 64y - 80 = 0$$

$$16y^2 - 64y - 9x^2 = 80$$

$$16(y^2 - 4y + 4) - 9(x^2) = 80 + 4(16)$$

$$16(y - 2)^2 - 9(x^2) = 144$$

$$\frac{16(y - 2)^2 - 9x^2 = 144}{144}$$

$$\frac{(y - 2)^2}{9} - \frac{x^2}{16} = 1$$

$$\frac{(y - 2)^2}{3^2} - \frac{x^2}{4^2} = 1$$

Thus, the center is $(0, 2)$.

SAMPLE PROBLEMS
RELATED TO
DIFFERENTIAL CALCULUS

GILLESANIA'S PRINCIPLE OF LIMITS

$$\text{Ex. 1: } \lim_{x \rightarrow 3} \frac{x^4 - 81}{x - 3}$$

Substitute $x = 2.9999$

$$\frac{(2.9999)^4 - 81}{(2.9999) - 3} = 107.99 \approx 108$$

$$\text{Ex. 2: } \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right)$$

Substitute $x = 0.0001$

$$\left(\frac{1}{0.0001} - \frac{1}{e^{0.0001} - 1} \right) = 0.49999 \approx 0.5$$

TRY TO SOLVE THIS

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 + 3x - 4}$$

Ans. 0.4

$$\lim_{x \rightarrow \infty} \frac{x + 4}{x - 4}$$

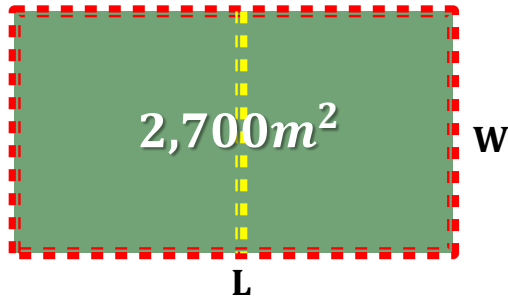
Ans. 1

EXAMPLE:

A rectangular field, having an area of $2,700 \text{ m}^2$, is to be enclosed by a fence, and an additional fence is to be used to divide the field down the middle. The cost of the fence down the middle is P12 per running meter, and the fence along the sides costs P18 per running meter. Find the dimensions of the field such that the cost of the fencing will be least.

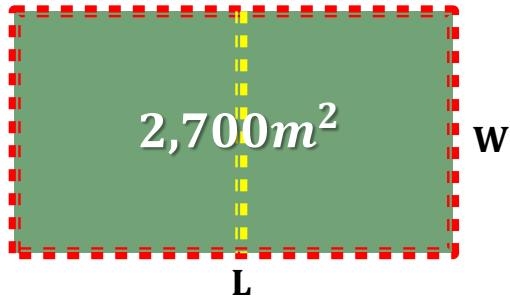
EXAMPLE:

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EXAMPLE:

A rectangular field, having an area of $2,700 \text{ m}^2$, is to be enclosed by a fence, and an additional fence is to be used to divide the field down the middle. The cost of the fence down the middle is P12 per running meter, and the fence along the sides costs P18 per running meter. Find the dimensions of the field such that the cost of the fencing will be least.



$$C = 18(2L + 2W) + 12W$$

$$C = 36L + 48W$$

$$L = \frac{2700}{W}$$

$$C = 36\left(\frac{2700}{W}\right) + 48W$$

$$C = \frac{97,200}{W} + 48W$$

$$C = \frac{97,200}{W} + 48W$$

$$\frac{dC}{dW} = -\frac{97,200}{W^2} + 48$$

$$-\frac{97,200}{W^2} + 48 = 0$$

$$W = 45$$

$$L = \frac{2700}{45} = 60$$

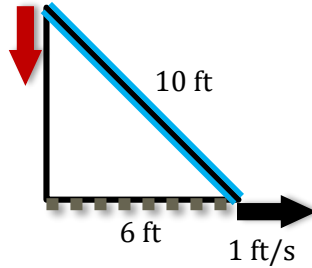
Thus, the dimensions of the field that minimize the cost of fencing are:

Length = 60 meters; Width = 45 meters

TRY TO SOLVE THIS

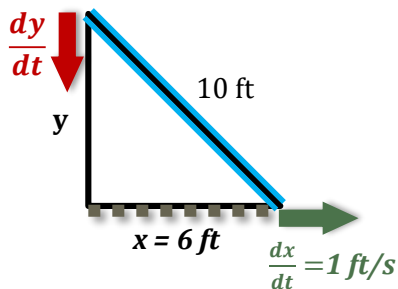
A ladder 10 feet long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate 1 ft/sec, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 feet from the wall?

- A. 0.75 ft/sec
- B. -0.75 ft/sec
- C. 0.5 ft/sec
- D. -0.5 ft/sec



A ladder 10 feet long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate 1 ft/sec, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 feet from the wall?

- A. 0.75 ft/sec
- B. -0.75 ft/sec**
- C. 0.5 ft/sec
- D. -0.5 ft/sec



By Pythagorean Theorem,

$$x^2 + y^2 = 10^2$$

Differentiating both sides wrt time,

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$x \frac{dx}{dt} + y \frac{dy}{dt} = 0$$

Getting the value of y:

$$x^2 + y^2 = 10^2$$

$$y = \sqrt{10^2 - 6^2} = 8 \text{ ft}$$

$$x \frac{dx}{dt} + y \frac{dy}{dt} = 0$$

$$6(1) + 8 \frac{dy}{dt} = 0$$

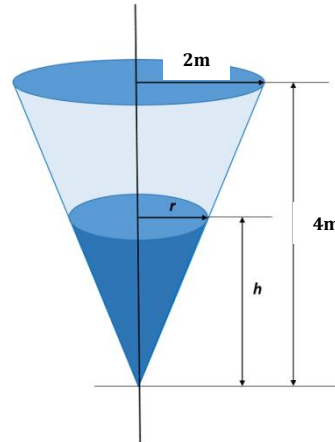
$$8 \frac{dy}{dt} = -6$$

$$\frac{dy}{dt} = -0.75 \text{ ft/s}$$

TRY TO SOLVE THIS

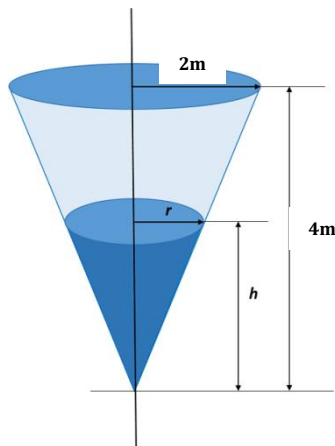
A water tank has the shape of an inverted circular cone with the base radius 2 meters and height 4 meters. If water is being pumped into the tank at a rate of 2 m/minute, find the rate at which the water level is rising when the water is 3 meters deep.

- A. -0.28/min
- B. 0.28 m/min
- C. 0.4m/min
- D. -0.4 m/min



A water tank has the shape of an inverted circular cone with the base radius 2 meters and height 4 meters. If water is being pumped into the tank at a rate of 2 m³/minute, find the rate at which the water level is rising when the water is 3 meters deep.

- A. -0.28/min
- B. 0.28 m/min**
- C. 0.4m/min
- D. -0.4 m/min



$$\frac{r}{h} = \frac{R}{H}$$

$$\frac{r}{h} = \frac{2}{4}$$

$$r = \frac{h}{2}$$

$$V = \frac{1}{3}\pi r^2 h$$

$$V = \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h$$

$$V = \frac{1}{12}\pi h^3$$

$$V = \frac{1}{12}\pi h^3$$

$$\frac{dV}{dt} = \left(\frac{1}{12}\pi\right)(3h^2)\frac{dh}{dt}$$

$$\frac{dV}{dt} = \frac{1}{4}\pi h^2 \frac{dh}{dt}$$

$$2 = \frac{1}{4}\pi(3)^2 \frac{dh}{dt}$$

$$\frac{dh}{dt} = 0.28 \text{ m/min}$$

TRY TO SOLVE THIS

The cost C of a product is a function of the quantity x of the product is given by the relation: $C(x) = x^2 - 4000x + 50$. Find the quantity for which the cost is minimum.

- A. 3000
- B. 2000
- C. 1000
- D. 1500

TRY TO SOLVE THIS

The cost C of a product is a function of the quantity x of the product is given by the relation: $C(x) = x^2 - 4000x + 50$. Find the quantity for which the cost is minimum.

A. 3000

B. 2000

C. 1000

D. 1500

$$C(x) = x^2 - 4000x + 50$$

$$\frac{dC}{dx} = 2x - 4000$$

$$2x - 4000 = 0$$

$$x = 2,000$$

TRY TO SOLVE THIS

Under a perfect competition, a firm can sell at a price of P200 per unit in all of a particular commodity it produces. If $C(x)$ pesos is the total cost of each day's production when x units are produced, and $C(x) = 2x^2 + 40x + 1400$, find the number of units that should be produced daily for the firm to have the greatest daily total profit.

- A. 30 units
- B. 35 units
- C. 40 units
- D. 45 units

TRY TO SOLVE THIS

Under a perfect competition, a firm can sell at a price of P200 per unit in all of a particular commodity it produces. If $C(x)$ pesos is the total cost of each day's production when x units are produced, and $C(x) = 2x^2 + 40x + 1400$, find the number of units that should be produced daily for the firm to have the greatest daily total profit.

- A. 30 units
- B. 35 units
- C. 40 units**
- D. 45 units

$$\text{Profit} = \text{Revenue} - \text{Cost}$$

$$P = 200x - (2x^2 + 40x + 1400)$$

$$P = -2x^2 + 160x - 1400$$

$$\frac{dP}{dx} = -4x + 160$$

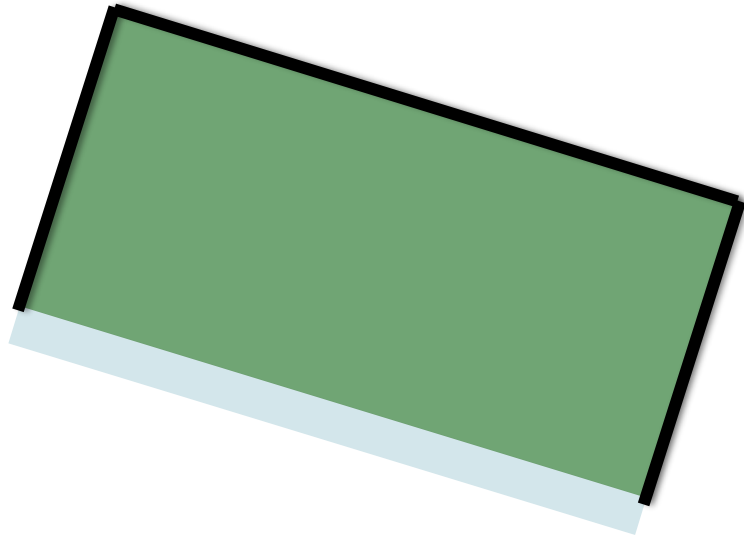
$$-4x + 160 = 0$$

$$x = 40$$

TRY TO SOLVE THIS

A farmer has 2,400 feet of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

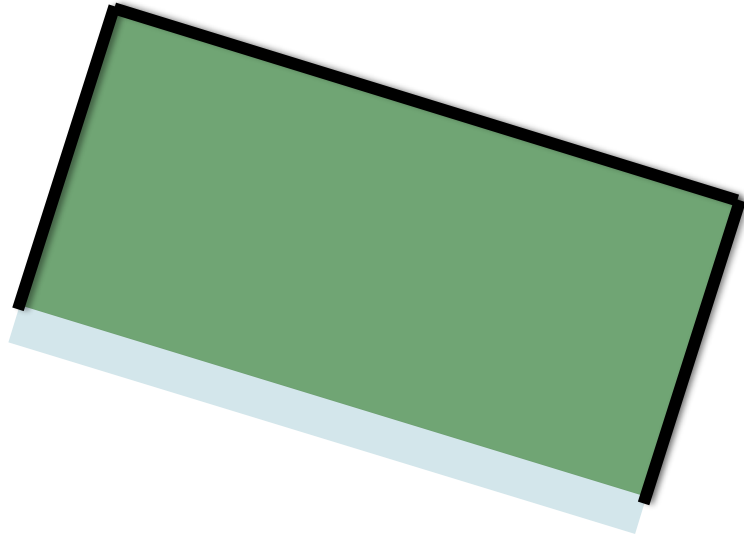
- A. 60 by 120
- B. 600 by 1,200
- C. 24 by 100
- D. 10 by 240



TRY TO SOLVE THIS

A farmer has 2,400 feet of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

- A. 60 by 120
- B. 600 by 1,200**
- C. 24 by 100
- D. 10 by 240





SAMPLE PROBLEMS
RELATED TO
INTEGRAL CALCULUS

Wallis Formula

$$\int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{[(m-1)(m-3) \dots (2 \text{ or } 1)][(n-1)(n-3) \dots (2 \text{ or } 1)]}{(m+n)(m+n-2) \dots (2 \text{ or } 1)} \alpha$$

where:

$\alpha = \frac{\pi}{2}$ when both m and n are even

$\alpha = 1$ if otherwise

m & n = positive integer, not equal to 1

EXAMPLES

Wallis Formula

$$\text{Ex. 1: } \int_0^{\frac{\pi}{2}} \sin^5 x dx = \frac{(4)(2)}{(5)(3)(1)} x \Big|_0^{\frac{\pi}{2}} = \frac{8}{15}$$

$$\text{Ex. 2: } \int_0^{\frac{\pi}{2}} \cos^8 x dx = \frac{(7)(5)(3)(1)}{(8)(6)(4)(2)} x \Big|_0^{\frac{\pi}{2}} = \frac{105\pi}{768}$$

$$\text{Ex. 3: } \int_0^{\frac{\pi}{2}} \sin^4 x \cos^5 x dx = \frac{[(3)(1)][(4)(2)]}{(9)(7)(5)(3)(1)} x \Big|_0^{\frac{\pi}{2}} = \frac{8}{315}$$

TRY TO SOLVE THIS

$$\int_0^{\frac{\pi}{2}} 12 \sin^5 x \cos^5 x dx$$

TRY TO SOLVE THIS

$$\int_0^{\frac{\pi}{2}} 12 \sin^5 x \cos^5 x dx = 12 \frac{[(4)(2)][(4)(2)]}{10(8)(6)(4)(2)} x \Big|_0^{\frac{\pi}{2}} = \mathbf{0.2}$$

TRY TO SOLVE THIS

$$\int_0^{\frac{\pi}{2}} 15 \sin^7 x dx$$

TRY TO SOLVE THIS

$$\int_0^{\frac{\pi}{2}} 15 \sin^7 x dx = 15 \frac{6(4)(2)}{7(5)(3)(1)} = \mathbf{6.857}$$

TRY TO SOLVE THIS

$$\int_0^{\frac{\pi}{2}} 3\sin^3 x dx$$

TRY TO SOLVE THIS

$$\int_0^{\frac{\pi}{2}} 3\sin^3 x dx = 3 \frac{2}{3(1)} (1) = 2$$

TRY TO SOLVE THIS

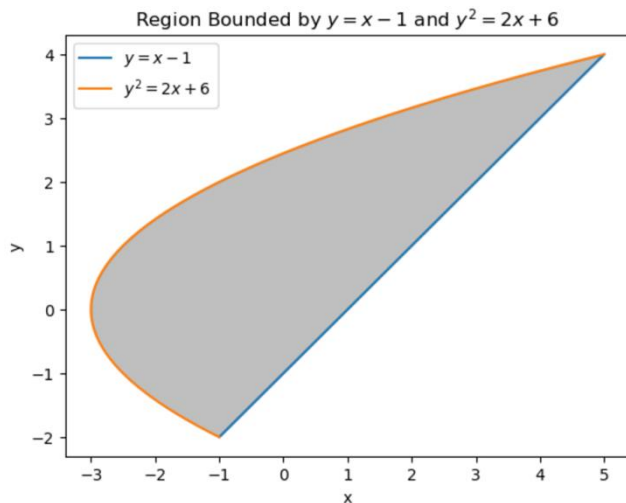
Find the area enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

- A. 15 sq units
- B. 16 sq units
- C. 17 sq units
- D. 18 sq units

TRY TO SOLVE THIS

Find the area enclosed by the line $y=x-1$ and the parabola $y^2 = 2x + 6$.

- A. 15 sq units
- B. 16 sq units
- C. 17 sq units
- D. 18 sq units**



Determining the upper and lower limits:

$$\begin{aligned}x &= y + 1 \\y^2 &= 2(y + 1) + 6 \\y^2 &= 2y + 2 + 6 \\y^2 - 2y - 8 &= 0 \\(y - 4)(y + 2) &= 0 \\y &= 4 \quad y = -2\end{aligned}$$

Determining the equations of line and parabola:

$$\begin{aligned}x &= y + 1 \text{ [equation of the line]} \\y^2 = 2x + 6 &\rightarrow x = \frac{y^2 - 6}{2} \text{ [equation of the parabola]}\end{aligned}$$

$$\int_{-2}^4 \left[(y + 1) - \left(\frac{y^2 - 6}{2} \right) \right] dy$$

[replace y with x in the calculator]

$$A = 18 \text{ sq. units}$$

TRY TO SOLVE THIS

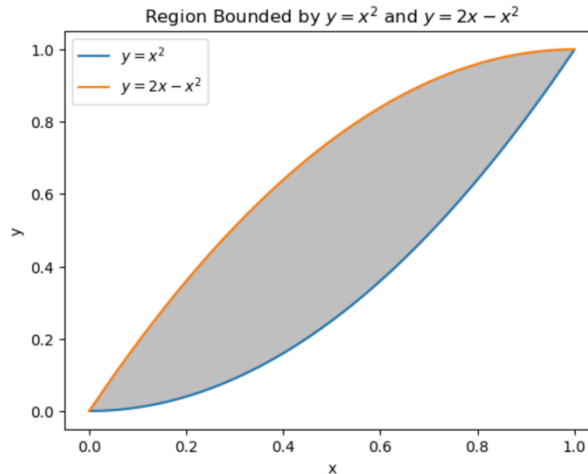
Find the area of the region bounded by $y = x^2$ and $y = 2x - x^2$

- A. $\frac{2}{3}$ sq unit
- B. $\frac{3}{8}$ sq unit
- C. $\frac{1}{3}$ sq unit
- D. $\frac{5}{8}$ sq unit

TRY TO SOLVE THIS

Find the area of the region bounded by $y = x^2$ and $y = 2x - x^2$

- A. $2/3$ sq unit
- B. $3/8$ sq unit
- C. $1/3$ sq unit**
- D. $5/8$ sq unit



Determining the upper and lower limits:

$$y = 2x - x^2$$

$$x^2 = 2x - x^2$$

$$2x^2 - 2x = 0$$

$$x = 1 \quad x = 0$$

Integral equation to find the area:

$$A = \int_0^1 [(2x - x^2) - x^2] dx$$

$$A = 1/3 \text{ sq. unit}$$

SAMPLE PROBLEMS
RELATED TO
DIFFERENTIAL EQUATIONS

TRY TO SOLVE THIS

Two months ago there were 5 mice, now there are 18. Assuming the growth continues like that, how many mice will there be two months from now?

- A. 44
- B. 65
- C. 98
- D. 123

TRY TO SOLVE THIS

Two months ago there were 5 mice, now there are 18. Assuming the growth continues like that, how many mice will there be two months from now?

- A. 44
- B. 65**
- C. 98
- D. 123

$$m(t) = \beta e^{kt}$$

At $t = 0$, $m = 5$ mice

$$\begin{aligned} 5 &= \beta e^{k(0)} \\ \beta &= 5 \end{aligned}$$

At $t = 2$ months, $m = 18$ mice

$$\begin{aligned} 18 &= 5e^{k(2)} \\ k &= 0.6404669 \end{aligned}$$

At $t = 4$ months,

$$\begin{aligned} m &= 5e^{(0.6404669)(4)} \\ m &\approx 65 \text{ mice} \end{aligned}$$

TRY TO SOLVE THIS

After 500 years, a sample of radium-226 has decayed to 80.4% of its original mass. Find the half-life of radium-226.

- A. 1590 years
- B. 1595 years
- C. 1598 years
- D. 2000 year

TRY TO SOLVE THIS

After 500 years, a sample of radium-226 has decayed to 80.4% of its original mass. Find the half-life of radium-226.

- A. **1590 years**
- B. 1595 years
- C. 1598 years
- D. 2000 year

$$m(t) = \beta e^{-kt}$$

Assume that original mass = 100g

At 500 years, $m = 80.4$ g

$$\begin{aligned} 80.4 &= (100)e^{-k(500)} \\ k &= 0.000436312 \end{aligned}$$

To find the half-life, say $m = 50$ g

$$50 = (100)e^{-(0.000436312)t}$$

$$\mathbf{t \approx 1,590 \text{ years}}$$

TRY TO SOLVE THIS

The mass of a radium 226 is given by $m(t) = \beta e^{-kt}$. Its half-life is 1,590 years. A sample of a radium-226 has a mass of 1,000 mg. Find the mass after 1,000 years.

- A. 650 mg
- B. 100 mg
- C. 165 mg
- D. 200 mg

TRY TO SOLVE THIS

The mass of a radium 226 is given by $m(t) = \beta e^{-kt}$. Its half-life is 1,590 years. A sample of a radium-226 has a mass of 1,000 mg. Find the mass after 1,000 years.

- A. **650 mg**
- B. 100 mg
- C. 165 mg
- D. 200 mg

$$m(t) = \beta e^{-kt}$$

Original mass = 1000 mg

So, at $t = 1,590$ years $m = 500$ mg

$$500 = (1000)e^{-k(1590)}$$

$$k = 0.000436$$

So, after 1000 years,

$$m(1000) = (1000)e^{-(0.000436)(1000)}$$

$$m(1000) \approx 650 \text{ mg}$$

TRY TO SOLVE THIS

Carbon 14, a radioactive isotope of carbon, has a half-life of 5,770 years. The amount of carbon present at any time t obeys the law $Q(t) = Q_0 e^{-kt}$. What is the decay constant k of Carbon 14?

- A. 0.00012
- B. -0.00012
- C. 0.5
- D. -0.5

TRY TO SOLVE THIS

Carbon 14, a radioactive isotope of carbon, has a half-life of 5,770 years. The amount of carbon present at any time t obeys the law $Q(t) = Q_0 e^{-kt}$. What is the decay constant k of Carbon 14?

- A. **0.00012**
- B. -0.00012
- C. 0.5
- D. -0.5

MISCELLANEOUS PROBLEMS

Three forces 20N, 30N, and 40N are in equilibrium. Find the angle between the 30-N and 40-N forces.

- a. $30^{\circ}, 15' 25''$ b. 28.96° c. 40° d. 25.97°

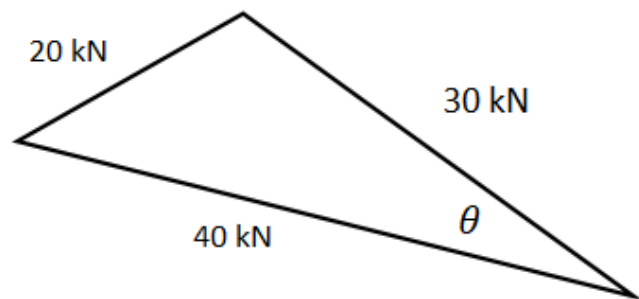
Solution:

Three forces are in equilibrium if their force vector forms a closed polygon (a triangle).

$$20^2 = 30^2 + 40^2 - 2(30)(40) \cos \theta$$

$$\cos \theta = 0.875$$

$$\theta = \mathbf{28.96^{\circ}}$$



The distance traveled by a train is given by the equation $x = 3t^2 + 2t + 4$ where x is the distance in kilometers and t is the time in hours. Determine the velocity of the train after traveling 10 kilometers.

a. 8.72 kph

b. 18.72 kph

c. 23.5 kph

d. 75.3 kph

Solution:

Distance, $x = 3t^2 + 2t + 4$

Velocity: $\frac{dx}{dt} = 6t + 2$

When $x = 10$

$$10 = 3t^2 + 2t + 4; 3t^2 + 2t + (-6) = 0$$

$$t = 1.12 \text{ hr.}$$

$$\text{Velocity} = \frac{dx}{dt} = 6(1.12) + 2$$

$$\text{Velocity} = \mathbf{8.72 \text{ kph}}$$

TRY TO SOLVE THIS

A student takes a part-time job paying a starting salary of 5,000 a month and is promised a fixed raise each month. How much is his monthly raise if he receives P7,000 on the eleventh month?

- A. P200
- B. P250
- C. P300
- D. P350

TRY TO SOLVE THIS

A student takes a part-time job paying a starting salary of 5,000 a month and is promised a fixed raise each month. How much is his monthly raise if he receives P7,000 on the eleventh month?

A. P200

B. P250

C. P300

D. P350

$$5000 + (10 x) = 7000$$

$$x = 200$$

TRY TO SOLVE THIS

A well driller charges P100 for the first 50 feet and P9.50 less for every 50 feet thereafter. How much would a 350-ft deep well cost?

- A. 500.50
- B. 552.50
- C. 643
- D. 700

TRY TO SOLVE THIS

A well driller charges ₱100 for the first 50 feet and ₱9.50 less for every 50 feet thereafter. How much would a 350-ft deep well cost?

- A. ₱500.50
- B. ₱552.50
- C. **₱643.00**
- D. ₱700.00

$$100 + ((100 - 9.50) \times 6) = \mathbf{643}$$

TRY TO SOLVE THIS

How much money do we have to invest at 3 percent compounded annually if we want to have 750 thousand pesos in the bank after ten years?

TRY TO SOLVE THIS

Ms. Sison invests part of P4,000 at 3% and the and the balance at 4% per annum . How much did she invest at each of these rates if she earns P135 in one year?

TRY TO SOLVE THIS

Daniel is constructing a ten - level tower. He puts ten blocks on the bottom level. If each level after that there is one block less than the level below it. The completed tower has how many blocks?

TRY TO SOLVE THIS

One computer can do a job twice as fast as the other. Working together, both computers can do the job in 2 hours. How long would it take each computer, working alone, to do the job?

TRY TO SOLVE THIS

The solution to the inequality $-10 < -2x < 20$ is

- A. -10 is less than x is less than 5
- B. 10 is greater than x is greater than -5
- C. -5 is greater than x is greater than -10
- D. -10 is less than x is less than -5

TRY TO SOLVE THIS

The solution to the inequality $-10 < -2x < 20$ is

- A. **-10 is less than x is less than 5**
- B. 10 is greater than x is greater than -5
- C. -5 is greater than x is greater than -10
- D. -10 is less than x is less than -5

TRY TO SOLVE THIS

A 10-meter tape is 5 mm short. What is the correct length in meters?

- A. 9.995 m
- B. 10.05 m
- C. 9.95 m
- D. 10.005 m

TRY TO SOLVE THIS

A 10-meter tape is 5 mm short. What is the correct length in meters?

A. 9.995 m

B. 10.05 m

C. 9.95 m

D. 10.005 m

$$10m - \left(5mm \times \frac{1m}{1000mm} \right) = 9.995m$$

TRY TO SOLVE THIS

The distance between two points measured with a steel tape was recorded as 916.58 ft. Later, the tape was checked and found out to be only 99.9 ft long. What is the true distance between the points?

- A. 935.66 ft
- B. 966.15 ft
- C. 955.66 ft
- D. 915.66 ft

TRY TO SOLVE THIS

The distance between two points measured with a steel tape was recorded as 916.58 ft. Later, the tape was checked and found out to be only 99.9 ft long. What is the true distance between the points?

- A. 935.66 ft
- B. 966.15 ft
- C. 955.66 ft
- D. 915.66 ft**

$$TD = MD + Error$$

$$\text{Length of tape used} = 100 \text{ ft}$$

$$\text{Error per tape length} = -0.1 \text{ ft}$$

$$\text{Measured Distance} = 916.58 \text{ ft}$$

$$TD = 916.58 \text{ ft} + (-0.1) \left(\frac{916.58}{100} \right) = 915.66 \text{ ft}$$

TRY TO SOLVE THIS

1° of Earth's rotation is equivalent to _____.

- a. 1 hour
- b. 4 minutes
- c. 30 minutes
- d. 360 seconds

TRY TO SOLVE THIS

1° of Earth's rotation is equivalent to _____.

- a. 1 hour
- b. **4 minutes**
- c. 30 minutes
- d. 360 seconds

Check this video for the review of identifying order and degree of differential equation:

<https://www.youtube.com/watch?v=m08Wge-a5k8>



*Thank you for listening
and wish you all the best!*

